

08INMO2

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Problem

Find all triples (p, x, y) such that $p^x = y^4 + 4$, where p is a prime and x and y are natural numbers.

Solution. Due to the Sophie Germain identity, we have: $y^4 + 4 = ((y - 1)^2 + 1)((y + 1)^2 + 1)$.

Let $(y - 1)^2 + 1 = p^a$ and $(y + 1)^2 + 1 = p^b$. We obviously have $a < b$.

$$p(y - 1)^2 + p \leq (y + 1)^2 + 1 \iff \left(\frac{p - 1}{2(p + 1)} \right) y^2 - y - 1 \leq 0. \text{ So, } \frac{1 - \sqrt{1 + \frac{2(p-1)}{p+1}}}{\frac{p-1}{p+1}} \leq y \leq \frac{1 + \sqrt{1 + \frac{2(p-1)}{p+1}}}{\frac{p-1}{p+1}}.$$

Since the lower bound decreasing and the upper bound increases as p increases, we get that $-2 < y < 2$. Since y is a natural number, we must have $y = 1$.

$y = 1 \implies p^x = 5 \implies p = 5, x = 1$. So the only pair is $\boxed{(p, x, y) = (5, 1, 1)}$.